

SUBSTITUTION TECHNIQUES

1. Ceaser Cipher

plain text - HELLO

key - 15

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$$E(P) = P + k \text{ mod } 26$$

$$E(H) = (7 + 15) \text{ mod } 26$$

$$= 22 \text{ mod } 26$$

$$= 22 = W$$

$$E(E) = (4 + 15) \text{ mod } 26$$

$$= 19 \text{ mod } 26$$

$$= T$$

$$E(L) = (11 + 15) \text{ mod } 26$$

$$= 26 \text{ mod } 26$$

$$= 0 = A$$

$$E(L) = (11 + 15) \text{ mod } 26$$

$$= 26 \text{ mod } 26$$

$$= 0 = A$$

$$E(O) = (14 + 15) \text{ mod } 26$$

$$= 29 \text{ mod } 26$$

$$= 3 = D$$

HELLO $\rightarrow$ WTAAD

2. Ceaser Cipher:

plain text - meet me after the party.

key - 3

$$E(P) = P + k \text{ mod } 26$$

$$E(m) = (12 + 3) \text{ mod } 26$$

$$= P$$

$$E(e) = (4 + 3) \text{ mod } 26$$

$$= h$$

$$E(e) = (4 + 3) \text{ mod } 26$$

$$= h$$

$$E(t) = (19 + 3) \text{ mod } 26$$

$$= w$$

$$E(m) = (12 + 3) \text{ mod } 26$$

$$= P$$

$$E(e) = (4 + 3) \text{ mod } 26$$

$$= h$$

$$E(a) = (0 + 3) \text{ mod } 26$$

$$= D$$

$$E(f) = (5 + 3) \text{ mod } 26$$

$$= I$$

$$E(t) = (19 + 3) \text{ mod } 26$$

$$= w$$

$$\begin{aligned}
 E(e) &= (4+3) \bmod 26 = 7 \\
 E(y) &= (17+3) \bmod 26 = 20 \\
 E(t) &= (19+3) \bmod 26 = 22 \\
 E(h) &= (7+3) \bmod 26 = 10 \\
 E(p) &= (15+3) \bmod 26 = 18 \\
 E(a) &= (0+3) \bmod 26 = 3 \\
 E(r) &= (17+3) \bmod 26 = 20 \\
 E(i) &= (8+3) \bmod 26 = 11 \\
 E(y) &= (24+3) \bmod 26 = 1
 \end{aligned}$$

cipher text - phbw ph diwlu wki sduwb.

3. Mono - alphabetic cipher

plain text - Cryptograph

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
P	F	R	A	I	T	Y	C	V	H	W	J	D	K	Z	L
Q	R	S	T	U	V	W	X	Y	Z						
S	E	G	M	X	B	N	Q	U	O						

cryptography $\rightarrow$ A E U L M Z Y E P L C U

4. play fair cipher

plain Text : Meet me Tomorrow keyword : keyword

K	E	Y	W	D
R	D	A	B	C
F	G	H	I/J	I
L	M	N	P	Q
S	T	U	V	

K E Y W D
 R D A B C
 F G H I J L
 M N P Q S
 T U V X Z

Me et me to mo rx ~~to~~ wx
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 nk ku nk kz ks BT kc bw

~~Me et me Tomorrow~~ → nkku nk kzksbtkcbw.

5. Polyalphabetic cipher

plain text: Active

keyword: Pascal.

P A S C A L
 15 0 18 20 11
 A C T I V E
 0 2 19 8 21 24

15 2 37 10 21 15
 P C L K V P

6. Hill cipher

plain text = HELP

$$\text{key} = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$E = kP \bmod 26$$

$$E = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 4 & e \\ 1 & p \end{bmatrix} \bmod 26$$

$$E = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 & 4 \\ 11 & 15 \end{bmatrix}$$

$$E = \begin{bmatrix} 54 & 57 \\ 69 & 83 \end{bmatrix}$$

$$E = \begin{bmatrix} 2 & 5 \\ 17 & 5 \end{bmatrix} = \begin{bmatrix} C & P \\ Y & P \end{bmatrix} = \text{CPYP}$$